Budhanilkantha School
Budhanilkantha
Kathmandu

April 23, 2022

The Teaching Committee and the Organizing Committee
Nepal Chemistry Olympiad
STEM Foundation Nepal

Dear Members of the Committee,

I had participated in the Nepal Chemistry Olympiad, round 1, held online via Zoom today on the 23rd of April. It has come to my awareness that despite myself selecting the correct option in all the questions, I was awarded merely 88 out of 100 and shown that I had three answers incorrect. However, I choose to refute this assertion that I scored 88 and am writing this letter justifying my claim.

Claim 1:

✕ The heat evolved when one mole of a strong acid and one mole of a strong base in dilute aqueous solution react is 0/4

- ○ 24.6 Kcal
- ◉ 12.3 Kcal ✕
- ○ dependent on the strong acid
- ○ dependent on both strong acid and strong base

Correct answer

- ◉ dependent on both strong acid and strong base

My answer for the above question is the correct answer.

Justification for Claim 1:

A strong acid by definition is an acid which ionizes completely in an aqueous solution. In the same manner, a strong base is a base which ionizes completely in an aqueous solution.

Taking HCl and NaOH as examples:

$$\mathrm{HCl\ (aq) + NaOH\ (aq) \rightleftharpoons NaCl\ (aq) + H_2O\ (l)}$$

Writing in ionic form and then canceling the spectator ions:

$$\mathrm{H^+(aq) + OH^-(aq) \rightleftharpoons H_2O\ (l)}$$

This indicates that the net reaction is the formation of H_2O.

Which means enthalpy change of neutralization for a strong acid and strong base is equivalent to the enthalpy change of formation of water from $H^+(aq)$ and $OH^-(aq)$.

The accepted value for this process is -13.48 ± 0.05 kcal for an $H_2SO_4 - NaOH$ system, and -13.52 ± 0.05 kcal for an $HCl - NaOH$ system (H. M. Papee, W. J. Canady, and K. J. Laidler. THE HEAT OF NEUTRALIZATION OF STRONG ACIDS AND BASES IN HIGHLY DILUTE AQUEOUS SOLUTIONS. Canadian Journal of Chemistry. 34(12): 1677-1682. https://doi.org/10.1139/v56-216).

Further, theoretically we can show using the van't Hoff equation:

$$\ln\left(\frac{K_{T2}}{K_{T1}}\right) = \frac{\Delta H^{\ominus}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Let us assume that the T initially that is $T_1 = 298K$, and finally it is $T_2 = 303K$.
At T_1, $K_{T1} = 1.008 \times 10^{-14}$, and at T_2, $K_{T2} = 1.471 \times 10^{-14}$.

$$\Delta H^{\ominus} = R\frac{\ln\left(\frac{K_{T2}}{K_{T1}}\right)}{\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

$$\Delta H^{\ominus} = 8.31\frac{\ln\left(\frac{1.471 \times 10^{-14}}{1.008 \times 10^{-14}}\right)}{\left(\frac{1}{298} - \frac{1}{303}\right)}$$

$$\Delta H^{\ominus} = -56722 \text{ J mol}^{-1}$$

$$\Delta H^{\ominus} = -56.7 \text{ kJ mol}^{-1}$$

$$\Delta H^{\ominus} = -13.6 \text{ kcal mol}^{-1}$$

As the van't Hoff plot is linear, we can say that for any value of T_1 and T_2 the enthalpy change will remain -13.6 kcal mol^{-1}.

This means that the heat evolved is numerically equal to the enthalpy of neutralization and hence around 13.6 kcal for the reaction between 1 mole of a strong acid and 1 mole of a strong acid. The options I was provided were none 13.6 kcal. However, 12.3 kcal was the closest option and hence I selected it.

Thus, I believe I should be awarded 4 marks for this question as my answer is correct.

Claim 2:

✕ Which of the following can be separated by sublimation method? 0/4

- ○ Anthracene
- ○ Paradichlorobenzene
- ○ Camphor
- ◉ All ✕

Correct answer

- ◉ Paradichlorobenzene

For this question, as well I claim that my answer is correct.

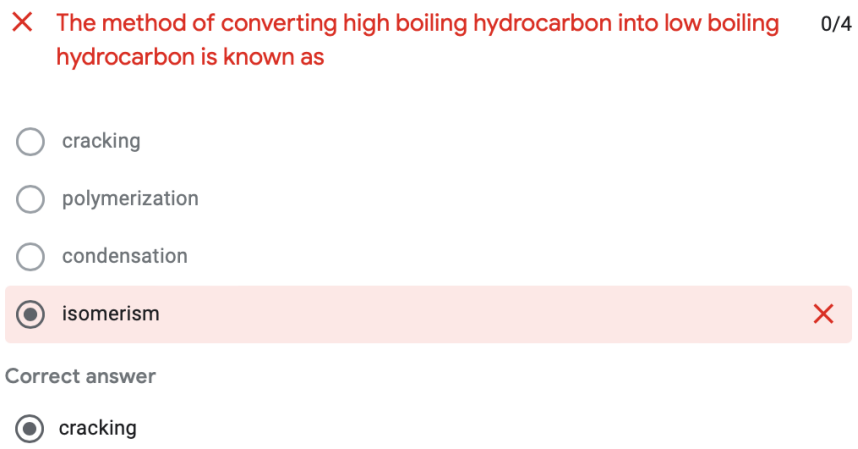
Justification for Claim 2:

According to the PubChem database of the National Center for Biotechnology Information of the National Library of Medicine of the National Institute of Health of the US, it is stated that Anthracene sublimes.
https://pubchem.ncbi.nlm.nih.gov/compound/Anthracene#section=Boiling-Point&fullscreen=true

Such is stated for p-dichlorobenzene too, as well as camphor.
https://pubchem.ncbi.nlm.nih.gov/compound/1_4-Dichlorobenzene#section=Decomposition
https://pubchem.ncbi.nlm.nih.gov/compound/Camphor#section=Boiling-Point&fullscreen=true

As such a credible source calls all three of them as a volatile, I believe the right option would be All (the option I have chosen), instead of p-dichlorobenzene as suggested by the answer key. And thus be awarded full for this question.

Claim 3:



I firmly believe that for this question both cracking and isomerism are correct answers.

Justification for claim 3:

It is true that cracking a hydrocarbon, will form hydrocarbons of lower boiling point. This is because the shorter the chain of hydrocarbon the lesser London dispersion forces It experiences. However, forming branched chain isomers has the same effect. This is because branching will decrease the surface area of the molecule and in turn decreasing the London dispersion forces it experiences.

If branching were less, there would be closer packing, that means more intermolecular forces take part in action and increase the boiling point. The opposite when branching is more. I believe with my explanation I should be provided full 4 marks on this question as well.

I believe that you have found my claims and justifications true and will in turn revise my score taking account these claims.

Sincerely,

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